

Cosmic-Ray Threshold Rates

Rate model

The CSV gives $F(E)$ in $(\text{m}^2 \text{sr s GeV})^{-1}$, with E in eV. These illustrative primary intensities are digitized from Sven Lafebre’s graphic (CC BY-SA 3.0). See `DATA_PROVENANCE.md` and `NOTICE` for attribution. Physical uncertainty is unquantified; the plotted energy convention is retained without resolving species or kinetic-versus-total energy distinctions.

For a one-sided horizontal surface of area A under an isotropic downward intensity, the rate in events per second is

$$R_{\text{CSV}}(> E_{\text{th}}; E_{\text{max}}) = \pi A_{\text{m}^2} 10^{-9} \int_{E_{\text{th}}}^{E_{\text{max}}} F(E) dE_{\text{eV}}.$$

Here $E_{\text{max}} \simeq 3.36 \times 10^{20}$ eV is the CSV endpoint. The integral uses power-law interpolation between tabulated points; 10^{-9} converts eV to GeV and π is the projected solid-angle factor. For $A = 1 \text{ km}^2$ the prefactor is $\pi \times 10^{-3}$.

The default omits flux beyond E_{max} . The optional `last-segment` model extrapolates the final slope $\gamma_{\text{last}} < -1$, adding

$$\int_{E_{\text{max}}}^{\infty} F(E) dE = \frac{F(E_{\text{max}})E_{\text{max}}}{-(\gamma_{\text{last}} + 1)}.$$

The tail is model dependent and matters near the upper endpoint. Atmospheric showers, detector response, solar modulation, and geomagnetic cutoffs are not modeled. A scalar aperture in $\text{m}^2 \text{sr}$ can replace πA .

Piecewise approximation

The fit covers 10^{10} – 10^{20} eV for the bundled CSV with truncation. Its five pieces share fitted breakpoint rates for continuity. Calibration minimizes squared log-rate residuals at the endpoints and logarithmic quarter points of each interval, counting shared endpoints once.

$$R_{\text{fit}}(> E) = R_i \left(\frac{E}{E_i} \right)^{-p_i}, \quad E_i \leq E < E_{i+1}.$$

E_i [eV]	E_{i+1} [eV]	R_i [$\text{s}^{-1} \text{km}^{-2}$]	p_i
10^{10}	4.5×10^{11}	3.898×10^8	1.516
4.5×10^{11}	6.4×10^{15}	1.216×10^6	1.672
6.4×10^{15}	6×10^{18}	1.382×10^{-1}	2.148
6×10^{18}	3.3×10^{19}	5.703×10^{-8}	1.880
3.3×10^{19}	10^{20}	2.313×10^{-9}	2.750

The final interval includes its upper endpoint. Table coefficients are rounded. For the unrounded fit, 10,001 thresholds uniformly spaced in $\log_{10} E$ give sampled absolute relative errors of 5.33% (median), 14.58% (95th percentile), and 17.03% (maximum), relative to the CSV reference.

Selected thresholds

Threshold E_{th} [eV]	Piecewise fit [$\text{s}^{-1} \text{km}^{-2}$]	CSV-truncated reference [$\text{s}^{-1} \text{km}^{-2}$]
10^{12}	3.198×10^5	3.298×10^5
10^{13}	6.805×10^3	6.031×10^3
10^{14}	1.448×10^2	1.519×10^2
10^{15}	3.081	3.474
10^{16}	5.3×10^{-2}	4.953×10^{-2}
10^{17}	3.767×10^{-4}	4.332×10^{-4}
10^{18}	2.678×10^{-6}	2.406×10^{-6}
10^{19}	2.183×10^{-8}	2.225×10^{-8}
10^{20}	1.097×10^{-10}	1.044×10^{-10}

Multiply the tabulated rates per km^2 by the area in km^2 to obtain events/s. Recalibrate the fit for another dataset or tail model.